

Curriculum Vitae

MANUEL KUHN

Date Prepared: 05/25/2026
Name: Manuel Kuhn
Office Address: Luisenstr. 57, 10117, Berlin, Germany
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Place of Birth: Dresden, Germany

Education:

10/2005 – 09/2007		Media Computer Science	Technical University of Dresden, Germany
10/2007 – 03/2013	Mag. rer. nat. (summa cum laude)	Psychology (<i>Advisors:</i> <i>Lukas Pezawas, MD</i> <i>Claus Lamm,</i> <i>PhD</i>)	University of Vienna, Austria
04/2013 – 08/2017	Dr. rer. biol. hum. (magna cum laude)	Human Biology (<i>Advisors:</i> <i>Tina B. Lonsdorf, PhD</i> <i>Christian Büchel,</i> <i>MD</i>)	Institute for Systems Neuroscience, Medical University Center Hamburg- Eppendorf, Germany

Postdoctoral Training:

09/2017 – 07/2019	Post-doctoral Associate	Research group Fear, Anxiety & Stress (<i>PI: Tina B. Lonsdorf, PhD</i>)	Institute for Systems Neuroscience, Medical University Center Hamburg- Eppendorf, Germany
08/2019 – 05/2020	Post-doctoral Associate	Affective & Translational Neuroscience Laboratory (<i>PI: Alexander J. Shackman,</i> <i>PhD</i>)	University of Maryland
06/2020 – 06/2022	Post-doctoral Fellow	Laboratory for Affective and Translational Neuroscience (<i>PI: Diego A. Pizzagalli, PhD</i>)	Center for Depression, Anxiety and Stress Research, McLean Hospital / Harvard Medical School

Faculty Academic Appointments:

07/2022 – Instructor, Department of Psychiatry, Harvard Medical School
11/2025

Appointments at Hospitals/Affiliated Institutions:

06/2020 – 06/2022	Research Fellow PhD	Laboratory for Affective and Translational Neuroscience (<i>PI: Diego A. Pizzagalli, PhD</i>)	Center for Depression, Anxiety and Stress Research, McLean Hospital / Harvard Medical School
07/2022 – 11/2025	Assistant Neuroscientist	Laboratory for Affective and Translational Neuroscience (<i>PI: Diego A. Pizzagalli, PhD</i>)	Center for Depression, Anxiety and Stress Research, McLean Hospital / Harvard Medical School
12/2025 - present	Senior Researcher	Institute of Medical Psychology (<i>PI: Christine Heim, PhD</i>)	Charité – Universitätsmedizin Berlin

Major Administrative Leadership Positions:

Local

2022 – Director, Neuroimaging and Instrumentation Core, Laboratory for Affective and
2025 Translational Neuroscience, McLean Hospital

Committee Service:

Local

2022 – Center for Depression, Anxiety and Stress Speaker Series, McLean Hospital, Chair
2025

Other Professional Positions:

2007 – 2012	Research Assistant	Vienna University of Technology
2009 – 2010	Research Assistant	University of Vienna
2011 – 2013	Research Assistant	Medical University of Vienna

Professional Societies:

2014	Organization for Human Brain Mapping (OHBM)	Student Member
2015 – present	The German Psychological Society (DGPs; Section: Biological Psychology and Neuropsychology)	Member
2015 – present	Research network for the (European) Interdisciplinary study of Fear and	Member

	Extinction Learning as well as the Return of Fear (EIFEL-ROF)	
2016	Society for Neuroscience (SfN)	Student Member
2024	Anxiety & Depression Association of America	Member
Grant Review Activities:		
2024	The European Region Tyrol-South Tyrol-Trentino's Interregional Research Projects (IRP) program, Reviewer	Austrian Science Fund (FWF)

Editorial Activities:

Ad hoc Reviewer

Behaviour Research and Therapy
Biological Psychiatry
Biological Psychiatry: CNNI
BJPsych Open
Cerebral Cortex
Cognitive, Affective, and Behavioral Neuroscience
European Psychiatry
International Journal of Psychophysiology
Human Brain Mapping
Nature – Communications
Nature – Mental Health
Nature – Scientific Reports
Nature – Translational Psychiatry
PLOS One
Psychophysiology
Psychological Medicine
Social, Cognitive and Affective Neuroscience

Honors and Prizes:

2010	Joint study program	University of Vienna / University of Urbana- Champaign, IL	Academic success
2011	Performance scholarship	University of Vienna	Academic success
2012	Performance scholarship	University of Vienna	Academic success
2015	Poster award	The German Psychological Society (DGPs)	Research
2017	Poster award	World Association for Stress Related and Anxiety Disorders	Research
2026	Charité Talentschmiede Award	Charité – Universitätsmedizin Berlin	Research

Report of Funded and Unfunded Projects

Completed

- 2023 – 2025 (start 07/23) A combined investigation of the protective impact of the NOF/Q antagonist LY-2940094 on stress-induced depression- and anxiety-related phenotypes in humans. Eric Dorris Memorial Fellowship (competitive local selection process, \$40,000).
Principal Investigator
This project elucidates the role of the nociceptin system on depression- and fear-related phenotypes within the same individual to determine a potential dissociable impact of a nociceptin antagonist on emotional disorders.
- 2024 – 2025 (start 04/24) A combined investigation of the protective impact of the NOF/Q antagonist LY-2940094 on stress-induced depression- and anxiety-related phenotypes in humans. NIMH Conte Center grant (P50 MH119467) Seed Grant Program (competitive selection process, \$41,000).
Principal Investigator
This project elucidates the role of the nociceptin system on depression- and fear-related phenotypes within the same individual to determine a potential dissociable impact of a nociceptin antagonist on emotional disorders.

Report of Local Teaching and Training

Charité – Universitätsmedizin Berlin (12/2025 – present); Teaching load: 4 SWS

- 2025 – 2026 Module M20: Mind and Pain as a Disease Model / seminars “Early Programming of Disease Vulnerability” and “The Stressed Brain – Psychoendocrine and Psychoimmunological Mechanisms of Stress-Related Disorders”; medical students (5th semester); 18 contact hours
- 2025/26 Module P03: The Human as a Biopsychosocial Being / seminars “Pain from a Psychological Perspective I” and “Pain from a Psychological Perspective II”; medical students (1st semester); 8 contact hours
- 2025/26 Module M07: Scientific Methods I / seminar “Assessment of Subjective Parameters in Human Medicine”; medical students (2nd semester); 6 contact hours
- 2025/26 Module B16: Health Literacy and Behavior Change / practical seminars “Practice of Health Psychology”; bachelor students (Health Sciences, 5th semester); 6 contact hours
- 2025/26 Module B02: Foundations of Social Sciences / seminar “Emotion”; bachelor students (Health Sciences, 1st semester); 4 contact hours
- 2026 Module B06: Health Psychology / seminar “Health-Related Behavior: Domains and Theories”; bachelor students (Health Sciences, 2nd semester); 4 contact hours
- 2026 Module M16: Sensory Organs / clinical integration teaching (KIT) “Counseling for Lifestyle Change”; medical students (4th semester); 3 contact hours

Report of Regional, National and International Invited Teaching and Presentations

- ☒ *No presentations below were sponsored by 3rd parties/outside entities*
- ☐ *Those presentations below sponsored by outside entities are so noted and the sponsor(s) is (are) identified.*

Local

- 2025 Linking Behavior to Neural Activity - Applications of Functional Magnetic Resonance Imaging and Computational Psychiatry / guest lecture
MIT, Cambridge, Massachusetts, USA

National

- 2013 SFB TRR-58 PhD Student Research Symposium / plenary talk
Marktbreit, Germany
- 2014 SFB TRR-58 PhD Student Research Symposium / plenary talk
Bad Gandersheim, Germany
- 2014 Annual SPM course / workshop host tutoring beginner and expert groups neuroimaging analyses using SPM
Institute for Systems Neuroscience
University Medical Center Hamburg-Eppendorf, Hamburg, Germany
- 2014 Annual MATLAB course / workshop host tutoring beginner groups MATLAB
Institute for Systems Neuroscience
University Medical Center Hamburg-Eppendorf, Hamburg, Germany
- 2015 Annual SPM course / workshop host tutoring beginner and expert groups neuroimaging analyses using SPM
Institute for Systems Neuroscience
University Medical Center Hamburg-Eppendorf, Hamburg, Germany
- 2015 Annual MATLAB course / workshop host tutoring beginner groups MATLAB
Institute for Systems Neuroscience
University Medical Center Hamburg-Eppendorf, Hamburg, Germany
- 2016 Annual SPM course / workshop host tutoring beginner and expert groups neuroimaging analyses using SPM
Institute for Systems Neuroscience
University Medical Center Hamburg-Eppendorf, Hamburg, Germany
- 2016 Annual MATLAB course / workshop host tutoring beginner groups MATLAB
Institute for Systems Neuroscience
University Medical Center Hamburg-Eppendorf, Hamburg, Germany
- 2016 Mismatch or Allostatic Load? Timing of Life-adversity Differentially Shapes Gray Matter Volume and Anxious Temperament / plenary talk
Psychologie & Gehirn 2016
Trier, Germany
- 2017 Annual SPM course / workshop host tutoring beginner and expert groups neuroimaging analyses using SPM
Institute for Systems Neuroscience
University Medical Center Hamburg-Eppendorf, Hamburg, Germany
- 2017 Annual MATLAB course / workshop host tutoring beginner groups MATLAB
Institute for Systems Neuroscience
University Medical Center Hamburg-Eppendorf, Hamburg, Germany
- 2017 The Impact of Life Events on the Return of Fear and Fear Generalization - Mechanisms and Clinical Implications / plenary talk
Symposium of the Clinical Psychology and Psychotherapy section of the DGPs

Chemnitz, Germany

- 2020 Life stress and the neurobiology of fear, anxiety, and affective temperament -
Investigations in healthy adults and their translational value for clinical research
Center for Depression, Anxiety, and Stress Research, Belmont, USA

International

- 2018 Introduction to Bayesian Analysis for Fear Conditioning Data / talk
European Meeting of Human Fear conditioning (EMHFC)
Cardiff, UK
- 2019 Introduction to Bayesian Analysis for Fear Conditioning Data / talk
European Meeting of Human Fear conditioning (EMHFC)
Würzburg, Germany
- 2021 Navigating the Multiverse of Skin Conductance Response Quantification Approaches /
plenary talk
Meeting Society for Psychophysiological Research (SPR)
Virtual meeting
- 2024 Computational Phenotyping of Effort-Based Decision Making in Unmedicated Adults
with Remitted Depression / plenary talk
49th Annual Conference "Psychology and Brain", Hamburg, Germany.
- 2025 Neural and Computational Phenotyping of Stress and Reward Processing in Remitted
Depression / invited talk
Decision Neuroscience of Human Interactions Laboratory, Heidelberg University,
Heidelberg, Germany.

Report of Education of Patients and Service to the Community

- ☒ *No presentations below were sponsored by 3rd parties/outside entities*
- ☐ *Those presentations below sponsored by outside entities are so noted and the sponsor(s) is (are) identified.*

Activities

- 2013 Biannual Night of Science Hamburg / Instructor
Hands-on experimental demonstrations and scientific dissemination to the public
- 2015 Biannual Night of Science Hamburg / Instructor
Hands-on experimental demonstrations and scientific dissemination to the public
- 2017 Biannual Night of Science Hamburg / Instructor
Hands-on experimental demonstrations and scientific dissemination to the public

Report of Scholarship

ORCID ID: 0000-0003-2210-9130

Peer-Reviewed Scholarship in print or other media:

* Indicates shared authorship

Research Investigations

1. Mertens G*, **Kuhn M***, Raes AK, Kalisch R, De Houwer J, Lonsdorf TB. Fear expression and return of fear following threat instruction with or without direct contingency experience. *Cogn Emot*. 2016 Aug;30(5):968-84. PMID: 25966279.
2. **Kuhn M**, Scharfenort R, Schümann D, Schiele MA, Münsterkötter AL, Deckert J, Domschke K, Haaker J, Kalisch R, Pauli P, Reif A, Romanos M, Zwanzger P, Lonsdorf TB. Mismatch or allostatic load? Timing of life adversity differentially shapes gray matter volume and anxious temperament. *Soc Cogn Affect Neurosci*. 2016 Apr;11(4):537-47. PMID: 26568620.
3. **Kuhn M**, Haaker J, Glotzbach-Schoon E, Schümann D, Andreatta M, Mechias ML, Raczka K, Gartmann N, Büchel C, Mühlberger A, Pauli P, Reif A, Kalisch R, Lonsdorf TB. Converging evidence for an impact of a functional NOS gene variation on anxiety-related processes. *Soc Cogn Affect Neurosci*. 2016 May;11(5):803-12. PMID: 26746182.
4. **Kuhn M***, Mertens G*, Lonsdorf TB. State anxiety modulates the return of fear. *Int J Psychophysiol*. 2016 Dec;110:194-199. PMID: 27496367.
5. Sjouwerman R, Niehaus J, **Kuhn M**, Lonsdorf TB. Don't startle me-Interference of startle probe presentations and intermittent ratings with fear acquisition. *Psychophysiology*. 2016 Dec;53(12):1889-1899. PMID: 27628268.
6. Lonsdorf TB, Menz MM, Andreatta M, Fullana MA, Golkar A, Haaker J, Heitland I, Hermann A, **Kuhn M**, Kruse O, Meir Drexler S, Meulders A, Nees F, Pittig A, Richter J, Römer S, Shibani Y, Schmitz A, Straube B, Vervliet B, Wendt J, Baas JMP, Merz CJ. Don't fear 'fear conditioning': Methodological considerations for the design and analysis of studies on human fear acquisition, extinction, and return of fear. *Neurosci Biobehav Rev*. 2017 Jun;77:247-285. PMID: 28263758.
7. Lucken U*, **Kuhn M***, Yang Y, Straube B, Kircher T, Wittchen HU, Pfleiderer B, Arolt V, Wittmann A, Ströhle A, Weber H, Reif A, Domschke K, Deckert J, Lonsdorf TB. Modulation of defensive reactivity by GLRB allelic variation: converging evidence from an intermediate phenotype approach. *Transl Psychiatry*. 2017 Sep 5;7(9):e1227. doi: 10.1038/tp.2017.186. PMID: 28872638.
8. Mertens G, Braem S, **Kuhn M**, Lonsdorf TB, van den Hout MA, Engelhard IM. Does US expectancy mediate the additive effects of CS-US pairings on contingency instructions? Results from subjective, psychophysiological and neural measures. *Behav Res Ther*. 2018 Nov;110:41-46. PMID: 30223152.
9. **Kuhn M**, Wendt J, Sjouwerman R, Büchel C, Hamm A, Lonsdorf TB. The neurofunctional basis of affective startle modulation in humans: Evidence from combined facial electromyography and functional magnetic resonance imaging. *Biol Psychiatry*. 2020 Mar 15;87(6):548-558. PMID: 31547934.
 - Early Career Investigator Commentary. Kaczurkin AN. Advances in mapping the startle eye-blink response onto neural circuits. *Biol Psychiatry*. 2020 Mar 15;87(6):e9-e11. PMID: 32081253.

10. Hur J, Smith JF, DeYoung KA, Anderson AS, Kuang J, Kim HC, Tillman RM, **Kuhn M**, Fox AS, Shackman AJ. Anxiety and the neurobiology of temporally uncertain threat anticipation. *J Neurosci*. 2020 Oct 7;40(41):7949-7964. PMID: 32958570.
11. Ehlers MR, Nold J, **Kuhn M**, Klingelhöfer-Jens M, Lonsdorf TB. Revisiting potential associations between brain morphology, fear acquisition and extinction through new data and a literature review. *Sci Rep*. 2020 Nov 16;10(1):19894. PMID: 33199738.
12. Hur J*, **Kuhn M***, Grogans SE, Anderson AS, Islam S, Kim HC, Tillman RM, Fox AS, Smith JF, DeYoung KA, Shackman AJ. Anxiety-Related Frontocortical Activity Is Associated With Dampened Stressor Reactivity in the Real World. *Psychol Sci*. 2022 Jun;33(6):906-924. doi: 10.1177/09567976211056635. Epub 2022 Jun 3. PMID: 35657777.
13. **Kuhn M**, Gerlicher AMV, Lonsdorf TB. Navigating the manyverse of skin conductance response quantification approaches - A direct comparison of trough-to-peak, baseline correction, and model-based approaches in Ledalab and PsPM. *Psychophysiology*. 2022 Sep;59(9):e14058. doi: 10.1111/psyp.14058. Epub 2022 Apr 2. PMID: 35365863.
14. Sjouwerman R, Illius S, **Kuhn M**, Lonsdorf TB. A data multiverse analysis investigating non-model based SCR quantification approaches. *Psychophysiology*. 2022 Jul 2:e14130. doi: 10.1111/psyp.14130. Epub 2022 Jul 2. PMID: 35780077.
15. Klingelhöfer-Jens M, Ehlers MR, **Kuhn M**, Keyaniyan V, Lonsdorf TB. Robust group- but limited individual-level (longitudinal) reliability and insights into cross-phases response prediction of conditioned fear. *Elife*. 2022 Sep 13;11:e78717. doi: 10.7554/eLife.78717. PMID: 36098500.
16. Koppold A, Kastrinogiannis A, **Kuhn M**, Lonsdorf TB. Watching with Argus Eyes: Characterization of emotional and physiological responding in adults exposed to childhood maltreatment and/or recent adversity. *Psychophysiology*. 2023 Jul;60(7):e14253. doi: 10.1111/psyp.14253. Epub 2023 Feb 2. PMID: 36727722.
17. Lewis MW, Webb CA, **Kuhn M**, Akman E, Jobson SA, Rosso IM. Predicting Fear Extinction in Posttraumatic Stress Disorder. *Brain Sci*. 2023 Jul 28; 13(8):1131. doi: 10.3390/brainsci13081131.
18. Ging-Jehli NR*, **Kuhn M***, Blank JM, Chanthrakumar P, Steinberger DC, Yu Z, Herrington TM, Dillon DG, Pizzagalli DA,*, Frank MJ*. Cognitive Signatures of Depressive and Anhedonic Symptoms and Affective States Using Computational Modeling and Neurocognitive Testing. *Biol Psychiatry Cogn Neurosci Neuroimaging*. 2024 Jul;9(7):726-736. doi: 10.1016/j.bpsc.2024.02.005. Epub 2024 Feb 23. PMID: 38401881.
19. Dillon DG, Belleau EL, Origlio J, McKee M, Jahan A, Meyer A, Souther MK, Brunner D, **Kuhn M**, Ang YS, Cusin C, Fava M, Pizzagalli DA. Using Drift Diffusion and RL Models to Disentangle Effects of Depression On Decision-Making vs. Learning in the Probabilistic Reward Task. *Comput Psychiatr*. 2024 May 3;8(1):46-69.
20. Grogans S, Hur J, Barstead MG, Anderson AS, Islam S, Kim HC, **Kuhn M**, Tillman RM, Fox AS, Smith JF*, DeYoung KA*, Shackman AJ*. Neuroticism/negative emotionality is associated

with increased reactivity to uncertain threat in the bed nucleus of the stria terminalis, not the amygdala. *J Neurosci*. 2024 Aug 7;44(32):e1868232024.

21. **Kuhn M***, Steinberger DC*, BendeZú JJ, Ironside M, Kang MS, Null KE, Brunner DL, Pizzagalli DA. Psychobiological Stress Response Profiles in Current and Remitted Depression: A Person-Centered, Multisystem Approach. *Biol Psychiatry Glob Open Sci*. 2024 Oct 9;5(1):100400. doi: 10.1016/j.bpsgos.2024.100400. PMID: 39582798.
 - Commentary. de Kloet ER, Molendijk ML. Stress System Concordance: A Signature of Resilience. *Biol Psychiatry Glob Open Sci*. 2025; 5:100427.
22. **Kuhn, M**, Palermo EH, Pagnier G, Blank JM, Steinberger DC, Long Y, Nowicki G, Cooper JA, Treadway MT, Frank MJ, Pizzagalli DA. Computational phenotyping of effort-based decision making in unmedicated adults with remitted depression. *Biol Psychiatry Cogn Neurosci Neuroimaging*. 2025 Jun;10(6):607-615. doi: 10.1016/j.bpsc.2025.02.006.
23. Cornwell BR, Didier PR, Grogans SE, Anderson AS, Islam A, Kim HC, **Kuhn M**, Tillman RM, Hur J, Scott ZS, Fox AS, DeYoung KA, Smith JF, Shackman AJ. A shared threat-anticipation circuit is dynamically engaged at different moments by certain and uncertain threat. *J Neurosci*. 2025 Apr 16;45(16):e2113242025. doi: 10.1523/JNEUROSCI.2113-24.2025.
24. Kim H, Park J, **Kuhn M**, Kim MJ, Hur J. Neuroticism modulates functional connectivity of the midcingulate cortex during emotional conflict. *Sci Rep*. 2025;15:13095. doi: 10.1038/s41598-025-97529-8..
25. Fullana M, Radua J, Savage H, Vilajosana E, Jamieson A, Abler B, Åhs F, Beckers T, Cardoner N, Cisler J, Diniz J, Bach D, Elsenbruch S, Greening S, Hoit D, Kaczurkin A, Keil A, Kindt M, Koch K, LaBar K, Lam C, Larson C, Lonsdorf T, Merz C, McLaughlin K, Neria Y, Pine D, van Reekum C, Shackman A, Soriano-Mas C, Spoormaker V, Stout D, Straube B, Straube T, Tuominen L, Visser R, Ahumada L, Arolt V, Batistuzzo M, Bazán P, Biggs E, Cano M, Chavarria-Elizondo P, Cooper S, Dannlowski U, De la Peña-Arteaga V, DeCross S, Domschke K, Ehlers M, Graner J, Hamm A, Herrmann MJ, Huggins A, Icenhour A, Juaneda-Seguí A, Junghofer M, Kircher T, Koelkebeck K, **Kuhn M**, Labrenz F, Lissek S, Lotze M, Lueken U, Margraf J, Martínez-Zalacáin I, Moeck R, Morriss J, Ortuño M, Pittig A, Porta-Casteras D, Richter J, Ridderbusch I, Rief W, Roesmann K, Rosén J, Rußmann A, Sjouwerman R, Spohrs J, Ströhle A, Suarez-Jimenez B, Ulrich M, Wittchen HU, Zhu X, Waller L, Walter H, Thompson P, Bas-Hoogendam JM, Groenewold N, Stein D, van der Wee N, Dunsmoor J, Marquand A, Harrison B. Neural correlates of human fear conditioning and sources of variability in 2199 individuals. *Nat Commun*. 2025;16:7870. doi: 10.1038/s41467-025-63078-x.
26. Woronko SE, Li M, Scott JN Jr, **Kuhn M**, Esfand SM, Bogdanov M, Boyle BW, Linton SR, Borchers LR, Zhukovsky P, Miller C, Bolton P, Li S, Meisner RC, Lees T*, Pizzagalli DA*. Alterations in aperiodic neural activity associated with major depressive disorder. *Nat Mental Health*. 2025;3(10):1181-1190. doi: 10.1038/s44220-025-00494-4.
27. Lees T*, Woronko SE*, Li M, Scott JN Jr, **Kuhn M**, Esfand SM, Bogdanov M, Boyle BW, Linton SR, Borchers LR, Zhukovsky P, Miller C, Bolton P, Li S, Meisner RC, Pizzagalli DA. Differences in high-frequency connectivity among large-scale functional networks linked to

major depressive disorder and treatment-resistant depression. *Biol Psychiatry Glob Open Sci.* 2025;5(6):100602. doi: 10.1016/j.bpsgos.2025.100602.

28. Koppold A, Lonsdorf TB, **Kuhn M**, Weymar M, Ventura-Bort C. Intraindividual Variability Matters for the Correspondence Between Subjective Arousal, Valence, and Physiological Responses. *Psychophysiology.* 2026;63(1):e70224. doi: 10.1111/psyp.70224.
29. Zhukovsky P*, **Kuhn M***, Borchers L*, Ren B, Woronko SE, Li M, Lam CST, Zhang EM, Ressler KJ, Brennan BP, Vitaliano G, Pizzagalli DA (in press). A Prospective Biomarker-guided Precision Medicine Trial of Bupropion and Sertraline for Major Depressive Disorder: Sequential Multiple Assignment Randomized Trial in Depression (SMART-D). *Nat Mental Health.*
30. Palermo EH, Fisher H, Blank JM, Anderson C, Bogdanov M, Steinberger DC, Long Y, Nowicki GP, Smith GW, Webb CA, **Kuhn M***, Pizzagalli DA* (in press). Characterizing emotion dynamics in remitted depression: A network approach using ecological momentary assessment. *J Mood Anxiety Disord.*
31. Didier PR, Grogans SE, Kaplan CM, Kim HC, Islam S, Anderson AS, Tillman RM, **Kuhn M**, Hur J, Fox AS, DeYoung KA, Smith JF, Shackman AJ (in press). Fear, anxiety, and the extended amygdala—Absence of evidence for strict functional segregation. *Soc Cogn Affect Neurosci.* 2026;nsag040. doi: 10.1093/scan/nsag040. PMID: 42234867.

Other peer-reviewed scholarship

1. **Kuhn M**, Popovic A, Pezawas L. Neuroplasticity and memory formation in major depressive disorder: an imaging genetics perspective on serotonin and BDNF. *Restor Neurol Neurosci.* 2014;32(1):25-49. PubMed PMID: 23603442.
2. Popovic A, **Kuhn M**, Pezawas L. [Chronic pain from a psychiatric perspective - Neurobiology and therapy]. *Manuelle Medizin.* 2013 February; 51(1):1-5. German.
3. Wendt J, **Kuhn M**, Hamm AO, Lonsdorf TB. Recent advances in studying brain-behavior interactions using functional imaging: The primary startle response pathway and its affective modulation in humans. *Psychophysiology.* 2023 Dec;60(12):e14364. doi: 10.1111/psyp.14364. PMID: 37402156.

Thesis:

1. **Kuhn M.** (Doctoral thesis) Fearing nature, nurture and thyself: Inter-individual differences shape the neurobiology of fear processing and anxious temperament. Universität Hamburg. 2017. <https://d-nb.info/1140603752>

Abstracts, Poster Presentations and Exhibits Presented at Professional Meetings (last 3 years):

1. Li M, **Kuhn M**, Woronko SE, Zhukovsky P, Borchers LR, Alexander MM, Landi M, Ren B, Vitaliano G, Brennan BP, & Pizzagalli DA. Using biological and behavioral markers to guide depression treatment in the Sequential Multiple Assignments for Randomized Trial (SMART): protocol overview. McLean Research Day 2023. Belmont, MA, USA.
2. Woronko SE, **Kuhn M**, Li M, Zhukovsky P, Borchers LR, Alexander MM, Landi M, Ren B, Vitaliano G, Brennan BP, & Pizzagalli DA. Using biological and behavioral markers to guide

depression treatment in the Sequential Multiple Assignments for Randomized Trial (SMART): randomization and data processing protocol. McLean Research Day 2023. Belmont, MA, USA.

3. Ging-Jehli NR, **Kuhn M**, Pedersen ML, Blank J, Herrington TM, Yu Z, Long Y, Thaikkandi S, Nowicki G, Steinberger D, Pizzagalli DA, Frank MJ. Linking behavioral and neural effects of acute stress in humans: a computational psychiatric approach with implications for depression. Winter Conference on Brain Research (WCBR) 2023. Snowbird, UT, USA
4. Grogans SE, Hur J, Barstead MG, Anderson AS, Islam S, Kim HC, **Kuhn M**, Tillman RM, Fox AS, Smith JF, DeYoung KA, Shackman AJ. Neuroticism is associated with increased reactivity to uncertain threat—but not ‘threat-related’ emotional faces—in the bed nucleus of the stria terminalis. Annual Meeting of the Society of Biological Psychiatry (SOBP) 2023. San Diego, CA, USA
5. Grogans SE, Hur J, **Kuhn M**, Anderson AS, Islam S, Kim HC, Tillman RM, Fox AS, Smith JF, DeYoung KA, Shackman AJ. Attenuated Safety-Related vmPFC Activation is Associated With Pervasively Elevated Real-World Distress. Annual Meeting of the Society of Biological Psychiatry (SOBP) 2023. San Diego, CA, USA.
6. Ging-Jehli NR, **Kuhn M**, Chanthrakumar P, Blank J, Steinberger D, Pizzagalli DA, Frank MJ. Cognitive signatures of anhedonia severity in a dynamic approach-avoidance conflict task with sequential sampling modeling. Computational Psychiatry Conference (CPCONF) 2023. Dublin, Ireland.
7. Falkenstein MJ, **Kuhn M**, Kuckertz JM, Woodson O, Bezaehler A, Webb C, & Beard C. Digital approach-avoidance training as a treatment augmentation for obsessive compulsive disorder. Flash talk presented by MJF at the Annual Meeting of the Society for Digital Mental Health. 2023. Virtual meeting.
8. Falkenstein MJ, Woodson O, **Kuhn, M**, Kuckertz JM, Bezaehler A, Webb C, & Beard C. Approach-Avoidance Training as an Augmentation of Exposure and Response Prevention for Contamination OCD. Symposium presented at the 2023 International Obsessive-Compulsive Disorder Foundation (IOCDF) Research Symposium, San Francisco, CA, USA.
9. Kim H, Park J, **Kuhn M**, Kim MJ, Hur J. Neuroticism and Brain Responses to Emotional Conflict. Society for Research in Psychopathology (SRP) 2023. St. Louis, MO, USA.
10. Koppold A, Kastrinogiannis A, **Kuhn M**, Ehlers M, Ruge J, Klingelhöfer-Jens M, Lonsdorf TB. Facing the Aftermath: Uncovering the Association of Early Life Adversity with Emotional Processing, Psychophysiology and Measurement Heterogeneity. Annual Meeting of the Society-for-Psychophysiological-Research (SPR) 2023. New Orleans, LA, USA.
11. Lam CT, Woronko SE, Li M, Borchers LR, **Kuhn M**, Zhukovsky P, Pizzagalli DA. Reward learning and clinical severity in treatment-seeking adults with major depressive disorder. McLean Research Day 2024. Boston, MA, USA.
12. Anderson C, Bogdanov M, **Kuhn M**, Palermo EH, Steinberger DC, Blank JM, Pizzagalli DA. The role of perceived stress and trait anticipatory pleasure in effort-based decision-making among individuals with remitted depression. McLean Research Day 2024. Boston, MA, USA.

13. **Kuhn M**, Palermo EH, Pagnier G, Blank JM, Steinberger DC, Long Y, Nowicki GP, Cooper JA, Treadway MT, Frank MJ, Pizzagalli DA. Computational Phenotyping of Effort-Based Decision Making in Unmedicated Adults with Remitted Depression. Anxiety & Depression Association of America Conference 2024. Boston, MA, USA.
14. Palermo EH, **Kuhn M**, Blank JM, Long Y, Nowicki GP, Webb CA, Pizzagalli DA. The association between emotional reactivity to an in-laboratory stressor and emotional variability in daily life varies based on depression history. Anxiety & Depression Association of America Conference 2024. Boston, MA, USA.
15. Anderson C, Bogdanov M, **Kuhn M**, Palermo EH, Steinberger DC, Blank JM, Pizzagalli DA. Perceived stress and early life adversity reduce motivation to exert physical effort for reward in healthy controls. Anxiety & Depression Association of America Conference 2024. Boston, MA, USA.
16. **Kuhn M**, Ging-Jehli NR, Palermo EH, Yu Z, Herrington TM, Blank JM, Thaikkandi S, Frank MJ, Pizzagalli DA. Neural correlates of stress-induced approach-avoidance behavior and conflict in remitted depression. Annual Meeting of the Society of Biological Psychiatry (SOBP) 2024. Austin, TX, USA.
17. Grogans SE, Hur J, Barstead MG, Anderson AS, Islam S, Kim HC, **Kuhn M**, Tillman RM, Fox AS, Smith JF, DeYoung KA, Shackman AJ. The neural systems and real-world mood dynamics underlying dispositional risk for internalizing illness. Annual Meeting of the Society of Biological Psychiatry (SOBP) 2024. Austin, TX, USA.
18. Zhukovsky P, Sneider J, **Kuhn M**, Woronko S, Mohan L, Borchers L, Ren B, Vitaliano G, Brennan B, Rubbani S, Coletti J, Ressler K, Pizzagalli DA. Nucleus Accumbens and Global Cortical Connectivity May Be Important Neural Correlates of Anhedonia in Major Depressive Disorder. Annual Meeting of the Society of Biological Psychiatry (SOBP) 2024. Austin, TX, USA.
19. Palermo EH, **Kuhn M**, Blank JM, Anderson C, Bogdanov M, Steinberger DC, Long Y, Nowicki G, Smith GW, Webb, CA, & Pizzagalli DA. Hippocampal Activation During an Approach/Avoidance Conflict Task is Associated with Approach/Avoidance Behavior in Daily Life. Society for Research on Psychopathology (SRP) 2024 Annual Convention, Montreal, Canada.
20. Smith GW, Bogdanov M, **Kuhn M**, Palermo EH, Anderson C, Steinberger DC, Blank JM, Pizzagalli DA. Physiological and Subjective Responses to a Social Stressor in Individuals with Depression, Remitted Depression, and Healthy Controls. Society for Research on Psychopathology (SRP) 2024 Annual Convention, Montreal, Canada.
21. Anderson C, Bogdanov M, **Kuhn M**, Palermo EH, Steinberger DC, Blank JM, Pizzagalli DA. The role of recent and early life stress in effort-based decision-making. Society for Research on Psychopathology (SRP) 2024 Annual Convention, Montreal, Canada.
22. Palermo EH, Fisher H, Blank JM, Anderson C, Bogdanov M, Steinberger DC, Long Y, Nowicki GP, Smith, GW, Webb, CA, **Kuhn, M***, & Pizzagalli DA*. Characterizing Emotional Dynamics in Remitted Depression: A network Approach Using Ecological Momentary Assessment. The

Special Interest Group Expo at the Association for Behavioral and Cognitive Therapies
58th Annual Convention, Philadelphia, PA, USA.

23. Castro GB, Didier P, **Kuhn M**, Kaplan CM, Hur J, Bradford DE, Curtin JJ, Smith JF, Shackman AJ. Alcohol Dampens Subjective, Psychophysiological, and Neurobiological Reactivity to Uncertain Threat. MiNDS Program at Temple University, Philadelphia, PA, USA.
24. Grogans SE, Hur J, Didier PR, Das S, Anderson AS, Islam S, Kim HC, **Kuhn M**, Fox AS, Smith JF, DeYoung KA, Shackman AJ. Increased reactivity to uncertain-threat anticipation in the bed nucleus of the stria terminalis and periaqueductal gray is associated with longitudinal worsening of internalizing symptoms, conditional on stress exposure. Annual Meeting of the Society of Biological Psychiatry (SOBP) 2025. Toronto, Canada.
25. Zhang E, Lam CST, Woronko SE, Li M, Bogdanov M, **Kuhn M**, Borchers LR, Zhukovsky P, Lees T, Pizzagalli DA. Error-Related Alpha Suppression in Major Depressive Disorder. Society for Research on Psychopathology (SRP) 2025 Annual Convention, Los Angeles, CA.
26. Lam CST, Zhang E, Woronko S, Li M, Borchers L, **Kuhn M**, Aruldass A, Sneider JT, Cohen-Gilbert J, Daskalakis NP, Ressler KJ, Zhukovsky P, Pizzagalli DA. Resting-State Functional Connectivity of the Nucleus Accumbens in Major Depressive Disorder: Associations with Anhedonia and Apathy Severity. Society for Research on Psychopathology (SRP) 2025 Annual Convention, Los Angeles, CA.
27. Woronko SE, Li M, Scott JN Jr, **Kuhn M**, Esfand SM, Bogdanov M, Boyle BW, Linton SR, Borchers LR, Zhukovsky P, Miller C, Bolton P, Li S, Meisner RC, Lees T, Pizzagalli DA. Alterations in Aperiodic Neural Activity Associated with Major Depressive Disorder. Society for Psychophysiological Research (SPR) 2025 Annual Meeting, Montréal, Canada.
28. Lees T, Woronko SE, Li M, Scott JN Jr, **Kuhn M**, Esfand SM, Bogdanov M, Boyle BW, Linton SR, Borchers LR, Zhukovsky P, Miller C, Bolton P, Li S, Meisner RC, Pizzagalli DA. Differences in High-Frequency Connectivity Among Large-scale Functional Networks Linked to Major Depressive Disorder and Treatment-resistant Depression. Society for Psychophysiological Research (SPR) 2026 Annual Meeting, Montréal, Canada
29. Nam K, Lam CST, Zhang EM, Woronko SE, Li M, **Kuhn M**, Bogdanov M, Borchers LR, Zhukovsky P, Thai M, Pizzagalli DA. *Differential reward learning in psychomotor subtypes of major depressive disorder*. McLean Research Day 2026, Belmont, MA, USA.
30. Palermo EH, Blank JM, Anderson C, Bogdanov M, Steinberger DC, Long Y, Nowicki GP, Smith GW, Chung Y, Rahimi-Eichi H, Baker JT, Webb CA, **Kuhn, M***, Pizzagalli DA*. *Increased negative affect and behavior are associated with decreased next day, but not same day, mobility in a sample of individuals with remitted depression and healthy controls*. P Anxiety and Depression Association of America (ADAA) 2026 Annual Conference, Chicago, Illinois, USA.
31. Kalinova K, Grogans SE, Didier PR, Anderson A, Islam S, Kim HC, **Kuhn M**, Tillman R, Hur J, Fox A, Smith JF, DeYoung KA, Shackman AJ. *Heightened Subcortical Reactivity to Threat and Exposure to Negative Life Events Jointly Predict the Longitudinal Course of Specific Internalizing Symptoms*. Annual Meeting of the Society of Biological Psychiatry (SOBP) 2026. New York, NY, USA.

32. Grogans S, DeYoung K, Hur J, Anderson A, Islam S, Kim HC, Wedlock J, Craig L, Tillman R, Das S, **Kuhn M**, Conway C, Fox A, Smith JF, Shackman AJ. *Understanding the Relevance of Dynamic Neural Responses to Anticipated Threat for the Emergence of Internalizing Illness*. Annual Meeting of the Society of Biological Psychiatry (SOBP) 2026. New York, NY, USA.